Kim Willcott

PhD Candidate • Physical Oceanography • Climate Dynamics

Irvine, CA kwillcot@uci.edu kwillcott LinkedIn

Research Profile

Physical oceanography PhD candidate developing reduced-order ocean circulation models to investigate the stability of the Atlantic and Southern Meridional Overturning Circulation under freshwater forcing. Research interests include climate dynamics, numerical modeling, dynamical systems, and ocean-ice interactions.

Education

University of California Irvine, PhD in Earth System Science — Sep 2024 – Present

- **Advisor:** Dr. Henri Drake
- **MSc in Earth System Science (en route), June 2026**
- **Coursework:** Geophysical Fluid Dynamics, Numerical Analysis, Physical Oceanography

University of British Columbia, MSc in Mathematics — Sep 2018 – Aug 2020

- **Advisor:** Dr. Neil Balmforth
- **Thesis:** Double diffusive convection and steady 2D salt finger solutions in porous media
- **Coursework:** Geophysical Fluid Dynamics, Differential Equations, Asymptotic Methods

University of British Columbia, BSc Hon. in Mathematics and Economics — Sep 2012 – May 2017

- 4.33/4.33 GPA
- **Coursework:** Number Theory, Abstract Algebra, Differential Equations, Econometrics

Experience

Graduate Teaching Assistant, University of California Irvine – Irvine, CA — Sep 2025 – Present

- Leading labs, hosting office hours, and grading assignments for ESS116 - Introduction to Environmental Data Science, taught in Python.

Graduate Student Researcher, University of California Irvine – Irvine, CA — Sep 2024 – Present

- Studying the stability of the overturning circulation in the presence of freshwater inputs. Specifically interested in the coupled dynamics between melting ice sheets and sea ice, and the overturning circulation.
- Using Python to implement models used to study stability.

Senior Pricing Data Analyst, GroupHEALTH – Surrey, BC — Oct 2022 – Jun 2024

- Built out new tools and processes to increase productivity and accuracy to handle the 1,000 quarterly renewing groups.
- Accurately priced group benefits using trends over the previous 5 years. Summarized and presented these results to VP of Underwriting and Director of Underwriting for further action.
- Trained new team members, and acted as an ongoing resource.
- Bridged the underwriting team and the in-house calculation software in terms of trouble-shooting and improving features.

Pricing Data Analyst, GroupHEALTH – Surrey, BC — Feb 2022 – Oct 2022

- Extracted data from SQL databases for pricing quarterly renewal rates, based on past usage and trends.
- Investigated claim trends and to address both internal questions from senior underwriters as well as external questions from clients and business partners.
- Reduced manual and repetitive work by creating Excel VBAs to efficiently update, print, and distribute packages for clients. By reducing manual processes, the opportunity for human error was also reduced, which increased efficiency in our processes.

Renewals Analyst, GroupHEALTH – Surrey, BC — Jan 2021 – Feb 2022

- Addressed clients' inquiries and requests, including but not limited to plan design changes as well as

cost-containment measures.

- Created usage and benchmarking reports via Excel and SQL for clients, providing valuable insight regarding their benefit plans.
- Accurately determined impacts of plan design changes based on claims experience, allowing clients to make informed decisions regarding their benefits packages.

Graduate Research Assistant, University of British Columbia – Vancouver, BC Sep 2018 – Aug 2020

- Studied double-diffusive processes in porous media using analytical and computational techniques and summarized results and findings using LaTeX in a clear and concise manner.
- Applied new conditions on an existing study, thereby changing the parameters and arriving at new results.
- Used MATLAB to help solve mathematical problems, such as solving for coefficients in evolution equations as well as plotting stability bounds.

Teaching Assistant, University of British Columbia – Vancouver, BC Sep 2018 – May 2020

- Tutored students on concepts such as ordinary and partial differential equations and calculus.
- Graded assignments, invigilated exams, and responded to student inquiries for MATH 256 (Differential Equations), MATH 257/316 (Partial Differential Equations).

Research Assistant, University of British Columbia – Kelowna, BC Sep 2017 – Jul 2018

- Worked on to two independent studies – Evolution of Predatory Models Under the Effects of Climate Change and Mutualistic Relationships Between Mycorrhizal Fungi and Plant Roots.
- Used MATLAB to create plots of phase planes in order to find climate-induced extinction.

Outreach and Mentorship

Supporting Women in Marine Sciences (SWMS) Sep 2025 - Present

- Acting as signing officer alongside two of my peers to start a new chapter at the University of California, Irvine. I will also be undertaking the role of treasurer for the group.
- The mission of this group is to promote visibility and celebrate women in marine sciences. This group will also serve as an opportunity to build community, provide professional development, and share experiences.

NexGen Mentor Program Sep 2025 - Present

- Mentoring students based on survey matching. This role allows me to offer insights regarding graduate school, as well as general guidance as needed.

FirstGen BEES Mar 2025

- Volunteered in leading lab tours for high school students, designed to introduce them to Earth and Environmental Science topics and spaces.

Presentations

Willcott, K., Drake, H., Sep 2026: "A 9-Box Overturning Circulation Model: Coupled Atlantic-Southern Ocean Dynamics" CalGFD, Los Angeles, CA. POSTER

Awards

School of Physical Sciences, C.A.R.E. Oct 2025 - Jun 2026

- $300 for the academic year
- Organized a weekly, focused writing time to support students and researchers, and to promote community-building within the School of Physical Sciences.

Faculty of Science Graduate Award 2018, 2019
Merit-based scholarship for academic excellence at a graduate level

- Award: $1,500/year

Deputy Vice-Chancellor Scholarship 2014, 2015, 2016
Merit-based scholarship for academic excellence at an undergraduate level

- Award: $1,000/year

Technologies

Programming Languages: Python, SQL, MATLAB, Excel VBA